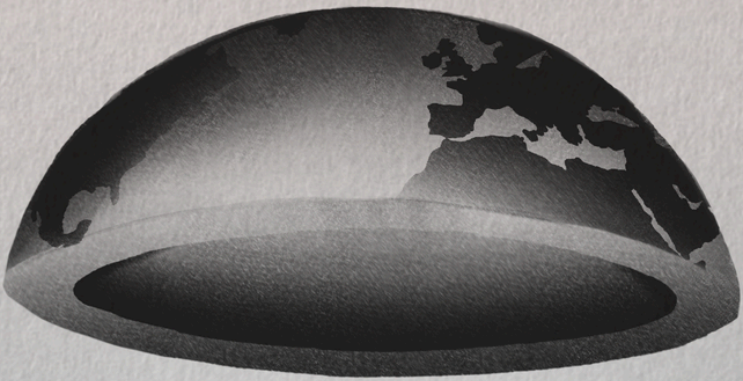
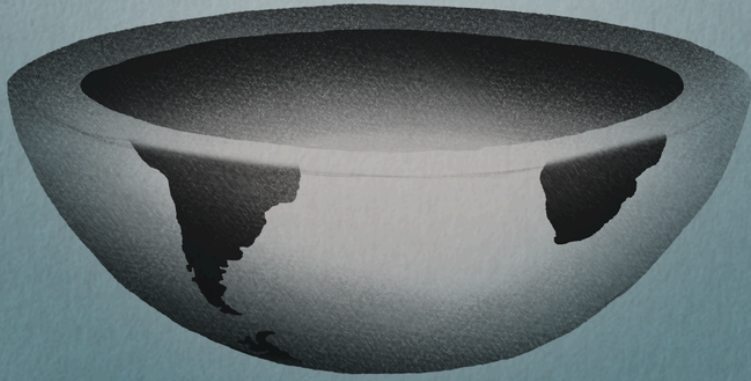




Rational



GOOD



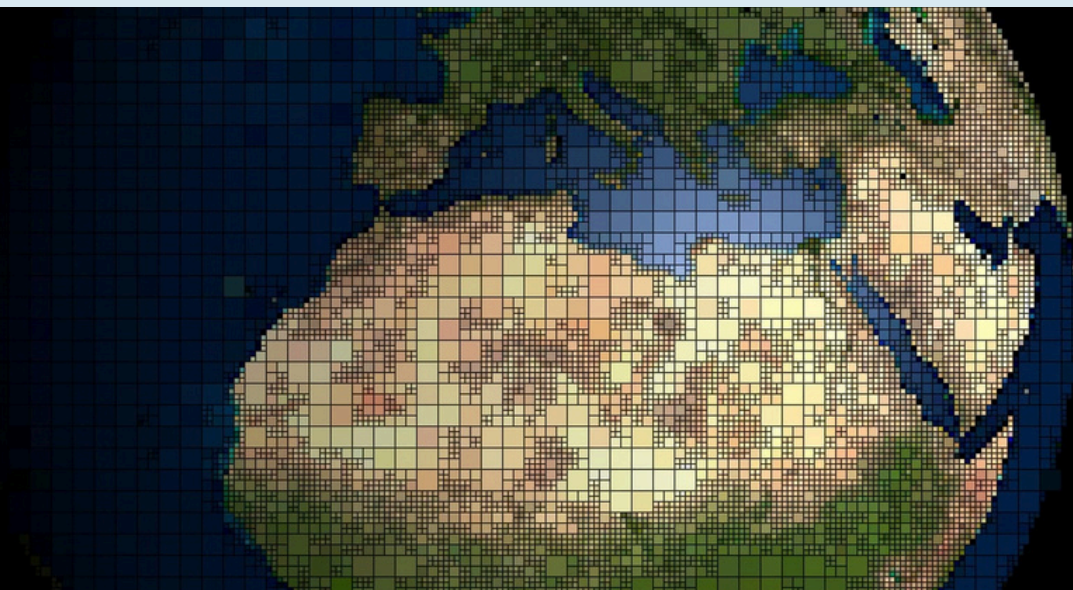
THE CENTRIFUGE

2026 October

TABLE OF CONTENTS



02	<u>ABOUT OUR THEME</u> Rational Good	24	<u>ADAPTATIVE BIOSECURITY PROGRAMME</u> Pandemic preparedness & the future of humanity
03	<u>OUR WINNERS</u> The top prizes of The Centrifuge Essay Competition		
05	<u>THE DOUBLE-EDGED BLADE OF MACHINE MATH:</u> How AI Solves Breakthroughs while Distorting Understanding	29	<u>SOLAR LATERNS AS AN OFF-GRID ENERGY SOLUTION</u> The net impact of novel lightning
11	<u>PILOTING A BROKEN ROCKET</u> Discovering what failed Apollo 13	34	<u>A SERENDIPITOUS SPILL</u> The Chemistry Behind Sweetness
20	<u>A PLAN FOR DEVELOPMENT, A PATH TO DESTRUCTION:</u> The Case of the Aral Sea	39	<u>OPPORTUNITIES PAGE</u> High-impact programs



THIS MONTH'S THEME - RATIONAL GOOD



Have you ever stopped to wonder what the single biggest problem in the world is? Perhaps your mind goes to the latest news headline, a chilling podcast, or familiar anxieties like climate change, digital burnout, and the looming threat of large-scale conflict.

Most of us worry about these issues, but we rarely calculate their exact odds or weigh their comparative severity. Instead, we leave those calculations to intuition and gut feeling. When we think about solving global challenges—whether through policy, innovation, or philanthropy—we rarely stop to ask whether our efforts or resources are being directed toward the interventions that actually do the most good.

After all, trying to help is inherently good, right?

Yet, in almost every other area of modern life, science has taught us that making sound decisions requires rigorous data, hypothesis testing, and critical evaluation. Why should doing good be any different?

This issue of *The Centrifuge* explores that exact question. We invited high school writers to share their perspectives on the role science plays in building a more effective, rational world. From the dual-edged sword of technological progress to the broad impacts of artificial intelligence, we have compiled the essays that stood out most for their perspicacity, creativity, and rigor. Our contributors tackled three central prompts:

- **The Trade-Off:** Many scientific breakthroughs solve one problem while creating another. Choose one technology and evaluate whether its overall impact is net positive.
- **The Grand Grant:** You have been given £100 million to improve humanity's future. You may fund only one scientific research program—which do you choose, and why?
- **The Road to Hell:** The world is full of good intentions and flawed implementations. Explore a project that sounded promising on paper but caused harm, and explain the science behind why it failed.



OUR WINNERS



1ST PRIZE - EKA KRISHNAN

U.S.A

THE ESSAY: A SERENDIPITOUS SPILL: THE CHEMISTRY BEHIND SWEETNESS.



2ND PRIZE - EDWARD CONNOLY

U.S.A

THE ESSAY: SOLAR LANTERNS AS AN OFF-GRID ENERGY SOLUTION



3RD PRIZE - DANQI ZHANG

U.K

THE ESSAY: ADAPTATIVE BIOSECURITY PROGRAMME



4TH PRIZE- NATAN T. NIGATU

ETHIOPIA

**THE ESSAY: A PLAN FOR DEVELOPMENT, A PATH
TO DESTRUCTION: THE ARAL SEA**



5TH PRIZE - IBRAHIM CHISHTI

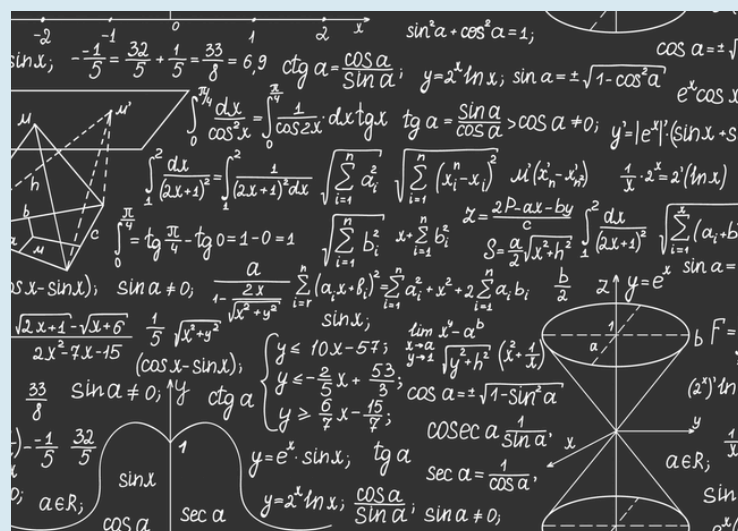
U.K

THE ESSAY: PILOTING A BROKEN ROCKET

SPECIAL MENTION - ANAY NAINA

U.K

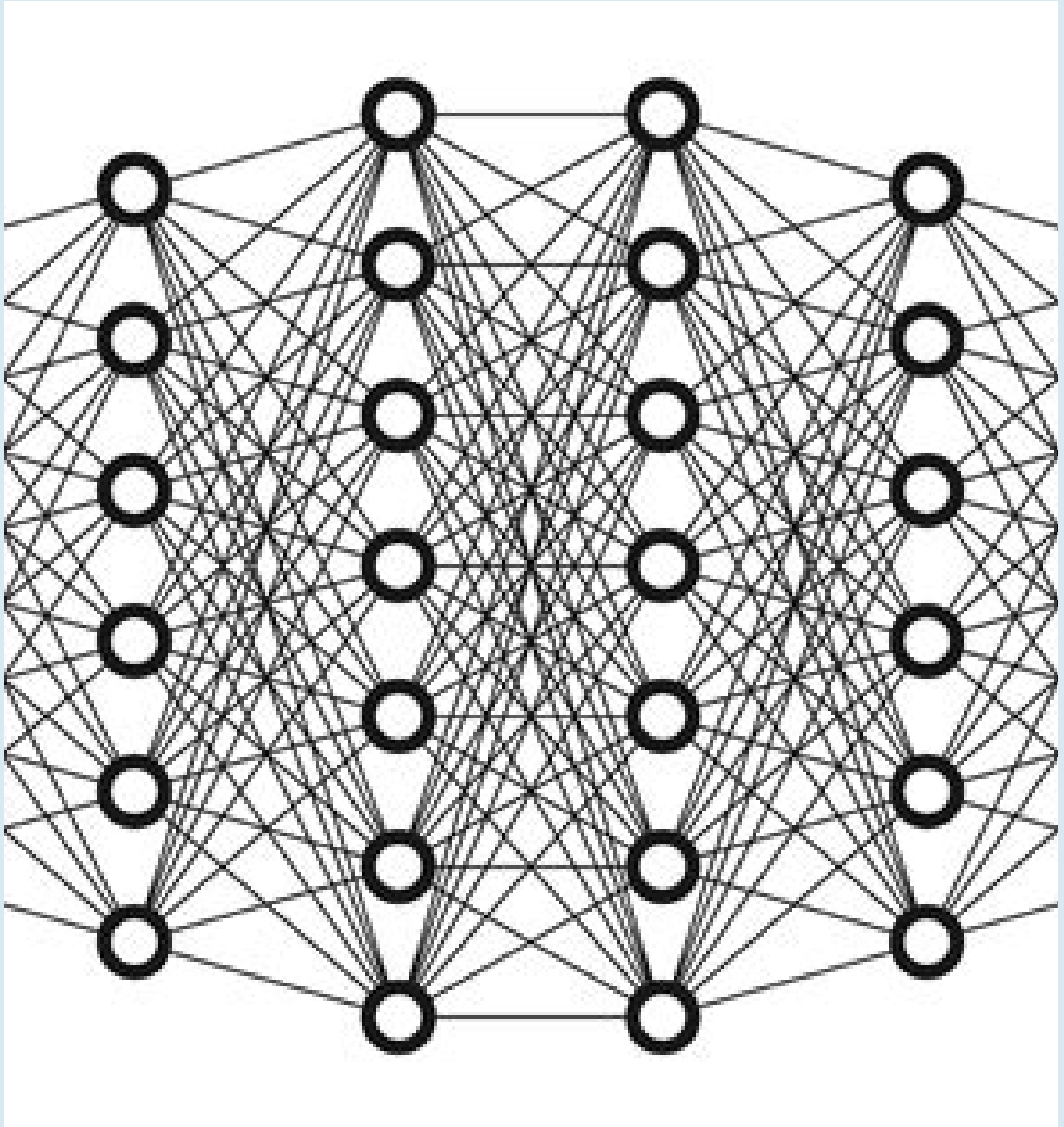
**THE ESSAY: THE DOUBLE-EDGED BLADE OF MACHINE
MATH: HOW AI SOLVES BREAKTHROUGHS WHILE
DISTORTING UNDERSTANDING**



THE DOUBLE-EDGED BLADE OF MACHINE MATH: **HOW AI SOLVES BREAKTHROUGHS WHILE** **DISTORTING UNDERSTANDING**



By Anay Naina - Runner Up



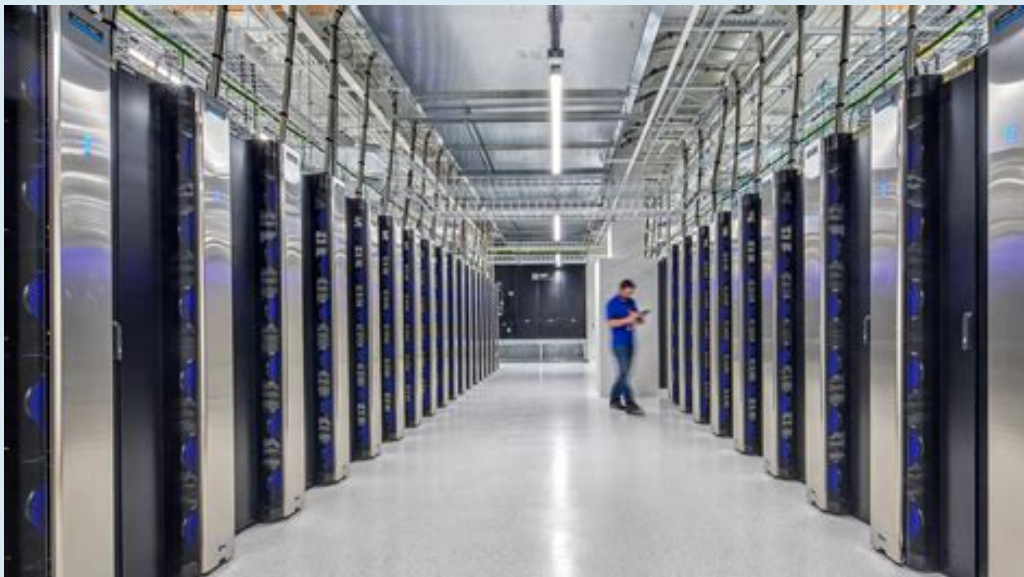


“A very central part of any mathematician’s life is this sense of connection to other minds, alive today and going back to Pythagoras. We are having this conversation with each other going over the millennia.”

–Steven Strogatz 2004

History demonstrates that the most profound scientific breakthroughs rarely offer clean solutions, but instead solve critical limitations of their era, and in turn uncover a new, often vastly more complex problem in their wake. For example, the Industrial Revolution led to a reduction in the need for manual labour through mechanisation, but brought about unprecedented environmental issues and degradation we are still grappling with today.

Today, Artificial Intelligence is driving a similar paradox within the mathematical and scientific community. Artificial Intelligence (AI) refers to computer systems designed to perform tasks that typically require human intelligence. These tasks include understanding spoken language, recognizing images, making decisions, and generating new content like text or art, or even solving complex problems. This ability, so humanlike yet so alien in the immense processing capabilities, has been a source of controversy within the mathematical community.





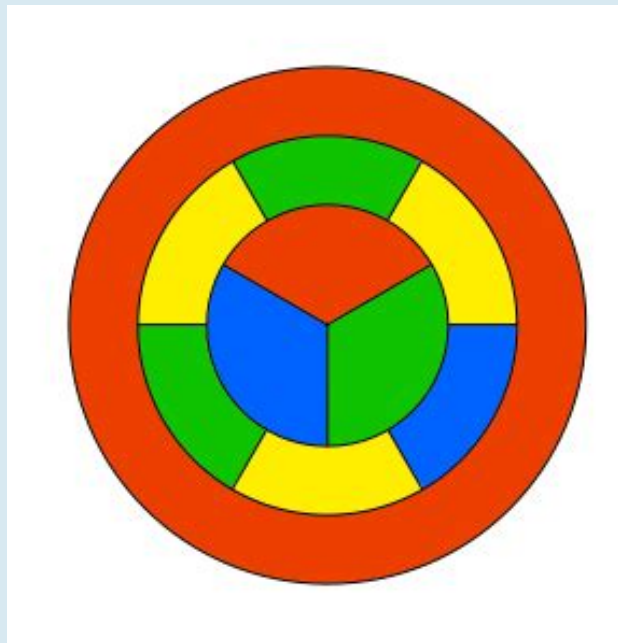
For centuries, mathematics has been defined by human comprehension, a pursuit of elegant, readable truths that could fit entirely within human working memory and be shared from one mind to another. Mathematics and the sciences are a community effort, a societal strive for understanding the universe. Institutions such as universities are built around the idea of “standing on the shoulders of giants”, highlighting the collaboration that is the foundation of mathematics and science. A noble, collective, human pursuit. However, just as all things human are, progress is potentially limited by our own biological limitations: we are unable to process hyper-dimensional complexity; analyzing systems, data, or spaces with thousands or millions of dimensions. AI overcomes this flaw of the human mind. Through immense computing power and thousands of agents, problems that would have taken humans years to think about can now be reasoned through in a matter of hours or even days.

However, this raises a significant philosophical problem known as the "black box.", meaning ideas that are too complex or abstract for humans to understand fully, or with speed.

“We now possess machines that can solve our most chaotic equations, yet we cannot understand the logic they use to do so.”

Despite the philosophical discomfort of losing human-readable understanding, the overall impact of AI on mathematics is overwhelmingly positive; it forces a necessary trade-off that allows humanity to solve existential, hyper-dimensional problems that would otherwise remain forever out of reach.

One such parallel from the past can be drawn to the historical controversy surrounding the Four Colour Theorem in 1976. When mathematicians Kenneth Appel and Wolfgang Haken set out to prove that any two-dimensional map could be coloured using only four colours without having adjacent regions share a shade, they ran into a problem of overwhelming scale. Rather than relying on a single elegant formula, they reduced the infinite possibilities down to 1482 distinct configurations. As manually checking thousands of complex map configurations exceeded human capabilities within a reasonable timeframe, Appel and Haken ran the exhaustive verification in a supercomputer.



An example illustration of the Four Color Theorem

This breakthrough marked a fundamental shift in scientific methodology, but it also provoked widespread skepticism within the mathematical community. For the first time, a major mathematical proof could not be verified by a single human mind scanning lines of equations on a chalkboard. Critics argued that relying on machine calculation fundamentally altered the discipline, raising a question almost suggestive of the classic “If a tree falls in the forest but no one is around to hear it, does it make a sound?”, as if a human cannot personally go through and understand every step of a proof, can humanity truly claim to “know” or “understand” the underlying truth?

The Four Colour Theorem’s solution demonstrates that computational assistance, while undeniably effective, threatens the traditional social contract of shared human understanding that Strogatz so succinctly highlights, the “conversation” between minds”. Despite these early doubts, the Four Colour Theorem merely involved a computer acting as a rapid calculator under explicit human instructions. Today’s AI paradigm has evolved far beyond basic computation, moving into autonomous problem-solving and deep neural network reasoning. As researchers attempt to peer into these high-dimensional systems, the tension between raw machine performance and human community and comprehension has only intensified, turning the initial concerns of 1976 into perhaps the defining scientific dilemma of the modern era.

This tension is clearest in recent AI breakthroughs involving the Navier-Stokes equations, one of the seven famous Millennium Prize Problems. These equations describe how fluids like air and water move, governing everything from climate patterns to turbulence around aeroplane wings. Because fluid motion is so chaotic, mathematicians have struggled to solve them for over two hundred years. Recently, swarms of AI agents operating autonomously have begun finding new patterns and potential solutions that humans missed. Unlike the 1976 supercomputer, which simply checked a list of maps built by humans, modern AI creates its own logical paths across high-dimensional spaces that human brains may struggle to visualise.



On September 8, 2026, OpenAI announced that a network of 10,000 autonomous AI agents running for 88 hours produced a 166-page proof claiming to solve the Navier-Stokes existence and smoothness problem by demonstrating the formation of a mathematical singularity, where the fluid reaches infinite speed in finite time with no energy change possible in the mathematical model, despite this not being able to occur in reality due to fluid viscosity, showing a critical breakdown in the model.

The rather unexpected release of Navier-Stokes generates a discussion about AI's role in mathematics; does the enormous benefit of simply solving problems outweigh the loss of the role of the journey to the solution?

Considering the positives brought about by AI, the immense data processing capabilities it suggests go beyond just visualisation, serving as an unprecedented "discovery engine" that shatters potential academic bottlenecks. Problems that have stumped human minds for centuries often require exploring millions of logical pathways and exhaustive edge cases. Autonomous AI agent swarms can systematically navigate these dead ends in days or hours rather than decades, identifying and structural solutions. Furthermore, when paired with formal proof assistants like Lean, AI guarantees absolute logical consistency by verifying mathematical steps line by line. Coupled with the immense speed of processing, AI offers a bright future for mathematics: Fermat's Last Theorem, which was predicted to take 5 years to translate into Lean by human means, was translated in 11 days by Anthropic's AI agents, accelerating the speed of human progress.

However, I argue that the value of a proof lies as much in the insight it provided, the "eureka!" moment of human comprehension, as it does in confirming that a statement is true. Whilst in the case of the Four Colour Theorem, where human understanding was critical in coming up with a method to eliminate all possible counterexamples, AI removes this work, this critical build up of tools and new mathematical techniques that are understood by the mathematical community when it runs the problem through its "black box". A problem is not truly solved when no one understands how the solution was brought about.

Furthermore, the vast scale of AI outputs creates a verification crisis. When an AI produces a proof spanning hundreds of thousands of lines of machine logic or Lean code, human peer review becomes nearly impossible without using another AI to verify the first. This reliance on machine-checking-machine risks creating a closed loop of trust, raising concerns about subtle software glitches, hidden systemic biases, or in the case of Lean, which requires code to compile to verify its truth, code filled with tautologies all for the purpose of achieving a proof.

Finally, I would argue that AI companies are currently using mathematical problems as simple benchmarks for their newest models, racing to complete problems, not to further mathematical understanding but to show off their raw computational capabilities and secure market dominance. This shifts the motivation of research from genuine intellectual curiosity to corporate branding.



When a model solves a centuries-old equation, the milestone is often packaged as a marketing showcase to attract investors and showcase processing power, rather than published as a collaborative tool designed for the broader academic community to learn from. Furthermore, this commercial race creates a rather unbalanced environment where elite tech companies hold the exclusive hardware resources needed to run multi-agent swarms, leaving public universities and independent researchers sidelined. If high-level problem-solving becomes concentrated behind proprietary corporate models, the open, democratic nature of scientific discovery is replaced by a closed ecosystem possibly driven by profit margins and proprietary benchmarks rather than the collective advancement of human knowledge.

In conclusion, I would argue that the rapid development of AI has solved many problems, increasing the speed at which mathematics and the sciences can progress, but just like the solution of the Four Colour Theorem, and the Industrial Revolution before it, brings about many problems that may cause significant damage to academia, including the potential privatisation of scientific progress, as well as the perhaps irreversible disruption of what I would argue is humanity's greatest treasure, the continuous stream of knowledge that each generation adds to and improves, as well as the loss of the tools to build upon the stream, tools that are gained through the journey of discovering mathematics, leading me to conclude that AI, a double-edged blade, has the potential to have a perhaps more overall negative impact on the sciences and mathematics.





PILOTING A BROKEN ROCKET: **DISCOVERING WHAT FAILED APOLLO 13**

By Ibrahim Chishti- 5TH place





Although initially a competition between NASA and the Soviet Space programme, after the successful landing of Apollo 11, the aims of the Apollo programme shifted to scientific advancement and exploration. Apollo 11 played it safe in the Sea of Tranquility for the sake of landing as quickly as possible. Once the aims shifted, Apollo 12 attempted a pinpoint landing next to the Surveyor 3 probe to study the effects of extended exposure to space on our probes. By the time Apollo 13 came round, the idea of going to the moon felt routine; there had been a mission more or less every 5 months, and people had lost the novelty of men on the moon. Apollo 13's Launch was televised by no major broadcasters. Insurance companies noted the difficulty, though, making it almost impossible for astronauts to get life insurance to provide for their families should they die in space.

All the same, the mission still went ahead as all missions had before them. The Saturn V rocket had two stages to get to Low Earth Orbit and a transfer stage to accelerate the ship to the Moon. The first stage was by far the most powerful vehicle built at the time, capable of 33 MN of thrust, equivalent to 27 million horsepower, burning through its 2 million kg of thrust within only 2.5 minutes, accelerating the craft up to 8 times the speed of sound, faster than any aeroplane.



The Apollo 13 Crew



3 hours into the flight, the 3rd stage completed the trans lunar injection, accelerating the ship enough to intercept the Moon in 3 days' time. So far, this mission had gone the same as those before. For those at mission control, it felt like just another workday. Jack Swigert, one of the astronauts on board, even remarked that he hadn't filed his taxes yet and asked how to do so from space. To everyone, it felt as though the ship was working as intended.

That was until 55 hours, 48 minutes into the flight, shortly after broadcasting on the TV about life in space, the mission transcript shows a broadcast on CAPCOM saying

"13, we've got one more item for you when you get a chance. We'd like you to stir up your cryo tanks."

A routine procedure for if the instruments were acting up, stirring up the cryo tanks (which hold fuel and oxidiser) would have just made them more uniform and allowed more accurate readings of remaining fuel. This time was different though:

CAP COM: Roger.

SC: Yeah, we got a main bus A under volt now to sh owin g.

CAP COM: Main A under volt.

SC: It's reading about 25 1/2, Main B is reading zip right now.

SC: (Garble)

CAPCOM: Standby one, Jim.

CAPCOM: 13, Houston, we'd like you to attempt to reconnect fuel cell 1 to Main A and fuel cell 3 to Main B.

Verify that quad delta is open.

SC: Okay, Houston, I'm showing, I tried to reset and fuel cell 1 and 3 are both showing gray flags.

SC: But they are both showing zip on the flows.

CAPCOM: We copy.

SC: Okay, Houston, are you still reading thirteen?



CAPCOM: That's affirmative. We're reading you. We're still trying to come up with some good ideas here for you.

SC: Okay, let me give you some readings (garble) in the interim to help Main A voltage Jack, got bus tie easy on it.

CAPCOM: Say again, Fred.

SC: In the interim to help out main A voltage, I've got Main bus tie bataC on. Or would you rather accept the 25 volts we are seeing on Main A.

CAPCOM: Okay, bus tie AC on. 13, Houston, we need OMNI CHARLIE, please.

SC: You got it?

CAPCOM: 13, Houston, we would like you to verify couple of readings for us. We would like the nitrogen pressure on fuel cell 1, we need the oxygen pressure on fuel cell 2.

SC: Okay. Nitrogen on 1 and oxygen on 2. Is that correct.

CAPCOM: Negative. Oxygen on 3. SC Okay. Okay. The systems test 1A, said zip. And 2 BAKER which is 3 oxygen says .6.

CAPCOM 2: BAKER says .6 and say again the other one.

SC: Fuel cell 1 nitrogen reads zero.

CAPCOM: Roger. Zero.

CAPCOM: 13, Houston. We'd like you to open circuit fuel cell 1, leave 2 and 3 as is.

SC: Okay. I'll get to work on that.

SC: And, Jack, our 02 cyro number 2 tank is reading zero, did you get that?

CAPCOM: 02 quantity number 2 is zero.



SC: That's AC, okay. Yes, that's AC and it looks to me looking out the hatch that we are venting something. We are venting something out into the - space.

CAPCOM: Roger, we copy your venting.

SC: It's a gas of some sort.

SC: Hey, fuel cell 1, you just wanted it off the line now, Jack, is that right?

CAPCOM: We just wanted you to open up fuel cell 1. SC Okay. She's off the line.

PAO: This is Apollo Control, Houston. This rapid exchange of conversation you've heard, may the main B bus is off the line, fuel cells 1 and 3 also off the line, fuel cell 2 is presently on the line. We now show 13 in an altitude of 178 643 nautical miles. We're at 56 hours, 12 minutes into the flight.

CAPCOM: 13, Houston, we see you getting close to gimbal lock there. We'd like you to bring up all quad C's on main A, quad C1, C2, C3, C4 on Main A and also bring B3 B4 up on Main A. 13, Houston, do you read?

SC: Yes. We got it. Affirm.

CAPCOM Okay. Can you tell us anything about the venting. Where it is coming from, what window you see it at -

SC It's coming out of window 1. Right now Jack, and could you give me the thrusters again?

END OF TAPE

The transcript you just read was 4 minutes after CAPCOM told the astronauts to stir the cryo tanks. This resulted in the instruments recording a drop in voltage in the Main A channel. This combined with a sudden jolt just after stirring was more than enough evidence that something had gone wrong. During the transcript the astronauts also realised that they were venting gas. The only gasses on board were fuel and oxidiser. They were venting fuel.

To get power on the Saturn V, the command module used a hydrogen fuel cell, taking a portion of the fuel and oxidiser to power the systems on board. Clearly they were now venting fuel and soon they would run out of power.

Luckily for them, Apollo 13 was planned to be a landing. They had the descent vehicle attached to the front of the ship (called the LM for Lunar Module) While the command module was in shambles, the LM was perfectly fine, as far as it was concerned it was still business as usual. An hour after the cryo tanks were stirred, the astronauts were instructed by CAPCOM to power down the CM and save the remaining fuel for re-entry. In the meantime they were to move to the LM and use it as a lifeboat of sorts. A craft built to support 2 people for a day now had to support 3 people for almost 5.



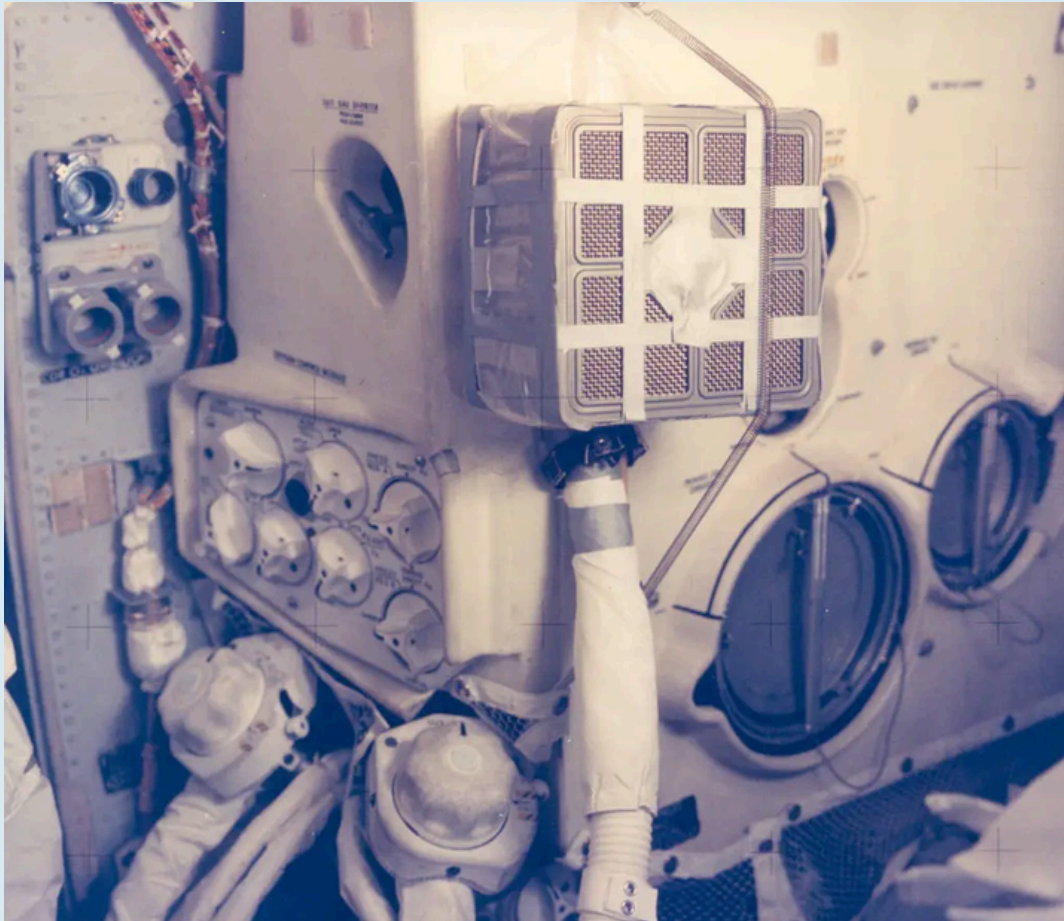
There was more though, if the crew wanted to get back home they would need to change from a trajectory going to the moon to a trajectory going back to the Earth. To do this they'd need an engine. The thing is the main engine was attached to the CM which had conveniently just been shut down. The later they fixed this trajectory the harder it would be to correct, Houston had to come up with a solution fast.

By this point it was clear that the Apollo 13 landing had to be scrapped. As such the only use for the LM right now was as a lifeboat for the crew. The way these missions were supposed to go was that the LM would detach from the CM in lunar orbit, then descend and land on the Moon. After surface operations were complete, the LM would then use its own engine to take off from the Moon and ascend back to the CM. This engine (along with the rest of the LM) was currently attached to the CM. Houston told the astronauts to use this to correct their course. All of the guidance and navigational data had to be transferred from the CM (which was now shut down) to the LM (who's guidance system hadn't begun yet because they were still a day away from the Moon). After instructions from Houston and hand-calculation by the astronauts on board, everyone racing to get the engine operational ASAP, the LM descent engine burnt for 34 seconds, just 5 hours after the initial incident. Now, all going well the crew would make it back to Earth in 4 days time.

All didn't go well.

Remember how I remarked that the LM would have to support 3 people for almost 5 days while only being designed for 2 people for about 20 hours? That was finally catching up to the astronauts. The CO2 scrubbers, lithium hydroxide canisters that react with CO2 in the air and allow the astronauts to breathe were running out. The LM scrubbers were almost completely saturated and it would soon become necessary to replace them. During construction, this eventuality was never planned for, no one thought to pack replacements because there was almost no scenario where the LM would be used for this long. The CM was supposed to be the one carrying the astronauts. Ideally you'd be able to put the CM scrubbers into the LM, but unfortunately the CM and LM were built by different teams on Earth, each following different standards. Namely the CM scrubbers were square while the housing for the LM scrubbers were circular. The CM scrubbers would not fit.

Luckily the guys at Houston were able to concoct a "mailbox" system, a way of attaching the square scrubber to the round hole using supplies on board (originally meant for lunar operations but now repurposed).



An interior view of the Apollo 13 Lunar Module and the "mailbox," a hack used to turn the lunar module into a lifeboat to get the crew safely back to Earth.

The next potential issue was water. The LM and CM used water from the fuel cells, however the CM had been shut down and the LM was running on as little power as possible, water was being produced at the slowest rate it possibly could. This could pose a huge problem, Houston calculated that, if the astronauts ration drinking water, the LM would run out of coolant water 5 hours before it was planned to detach. This wouldn't be an issue however, previous missions (i.e. Apollo 8-12) had shown that the LM could operate 7-8 hours after water cutoff.

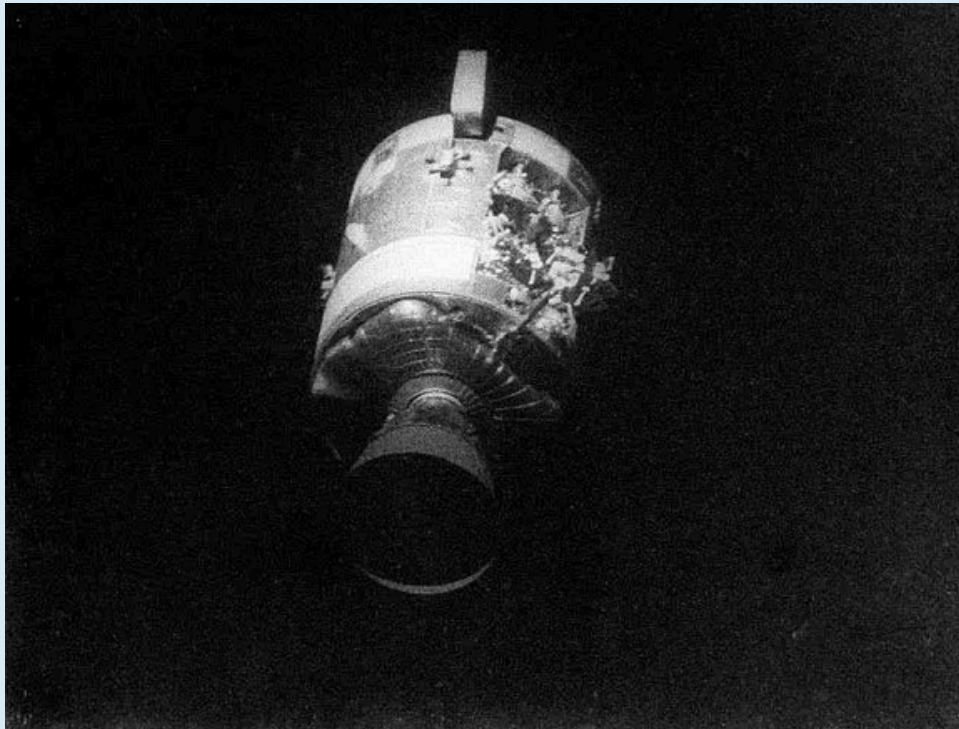
To save on power the astronauts also turned off all heaters, now it was getting to them. The LM had decreased down to only 3 degrees Celsius, water condensed on the walls and electronics (though it didn't cause any electrical issues because the electronics were well insulated from Apollo 1). Despite this the astronauts voiced no complaints.

133 hours into the flight, re-entry preparations had to begin. The CM and navigational computer had to be turned back on after a complete shutdown. Furthermore it had to be done with the reduced power output of the fuel cell. Tests on the ground were run for the past few days to see the optimal way to power on and the sequence was relayed to the crew. After startup it was clear that there was an issue. All the computers on the ground had been operating under the assumption that the crew would have brought back a couple of tonnes of moon rocks with them. Since they never landed, they were underweight and their trajectory was slightly off.



When re-entering the Earth's atmosphere, if you come in too shallow, the air won't brake you enough and you'll crash and die on the ground. If too shallow, the atmosphere will act like it does over a plane wing, and just send you back out to space again. Apollo 13 had to do another burn, but the LM navigational computer wasn't made to work so close to Earth, the astronauts would have to navigate themselves. Houston told the astronauts to burn for 14 seconds, using the terminator line of Earth as a reference. After completing this burn the astronauts were finally on the correct trajectory. They were set to splashdown in the South Pacific Ocean.

Half an hour later, the Service Module was jettisoned, allowing the astronauts to see the damage for the first time.



This view of the severely damaged Apollo 13 Service Module (SM) was photographed from the Lunar Module/Command Module (LM/CM) following SM jettisoning. As seen in this cropped image, enlarged to provide a close-up view of the damaged area, an entire panel on the SM was blown away by the apparent explosion of oxygen tank number two.

There was one more issue to consider, though there was still no way to overcome it. What if the heat shield was damaged? In that case the astronauts would burn up in the atmosphere. During re-entry, bleeding off so much speed causes the air around the spacecraft to compress into a plasma. This releases a lot of EM waves, enough to make communication by radio effectively impossible. This causes a blackout for any re-entering craft that lasts about 7 minutes. As the crew entered the Earth's atmosphere, they too entered the 7 minutes of silence.

All Houston could do was wait



After 7 minutes the crew emerged again, there was much celebration. Many of the people on the ground had not gone home since the explosion, sleeping under their desks and working from when they woke up to when they went to sleep to get these astronauts home. They had done it.

The astronauts were in sorry shape after the ordeal, together they had lost 15kg during the mission and were severely dehydrated. Fred Haise had developed a urinary tract infection due to a combination of radiation, weightlessness and dehydration.

Later investigations found that a poorly insulated wire was next to one of the Oxygen tanks. All it needed was a disturbance to ignite some of the fuel and cause the explosion, the disturbance being the stirring of the Oxygen tanks 56 hours in. Later missions carried out the experiments that were meant to be carried out by Apollo 13, no scientific knowledge was lost in the long term.

On later missions the CO₂ scrubbers were standardised and the trajectory taken became a free return trajectory, meaning that anything going wrong would send the crew straight back home.

None of the crew ever flew in space again, Jim Lovel, the commander, retired in 1973 and passed away in August 2025. Fred Haise, after recovering from his urinary tract infection, was planned to go on a later Apollo mission which was unfortunately cancelled due to budget cuts from Congress, he went on to work on the space shuttle programme before leaving NASA to pursue Aerospace full time, currently the only living crew-member as of September 2026. Jack Swigert left NASA in 1977, going into politics and later elected to join the House of Representatives but passing away from bone cancer in 1982 before he went into office.

Apollo missions continued until the last one in 1972, Apollo 17 (the first to have an actual scientist). After that NASA's human spaceflight department pivoted to low Earth orbit operations and built the Space Shuttle, playing a pivotal role in the construction of the ISS.

A film adaptation of the mission was later made and a fanciful term was coined for the mission.

“The Successful Failure”



A PLAN FOR DEVELOPMENT, A PATH TO DESTRUCTION: THE EXAMPLE OF THE ARAL SEA



By Natan Tibebu Nigato – 4TH place





Have you ever heard a quotation that says, “Resources are limited, but creativity is unlimited”? Well, this is true because there are many problems that have either been solved or are waiting for a solution. That is, people always think about how to tackle problems using their unlimited creative potential to create a way to stop suffering from a certain problem in their community. However, these proposed solutions may not always hit their target. They might result in another huge consequence, one that is even greater than the problem itself. That is what happened to the Aral Sea.

With an area of 68,000 km² (26,300 sq mi), it was formerly declared the third-largest lake in the world, found in the Aktobe and Kyzylorda regions of Kazakhstan and the Karakalpakstan autonomous region of Uzbekistan. Before it started decreasing in area, it was known for its low salinity, which was 10 grams per litre, even less than most seawater in the world. But after the 1960s, this whole story was totally reversed.

In earlier times, Central Asia was dry and arid. Because of this, rainfall was insufficient to support large-scale cultivation of water-intensive crops such as cotton. There were only two major rivers in the region, known as the Amu Darya and Syr Darya, which supplied the Aral Sea. Then, after the Soviet Union started to take control of Central Asia, the government started to see the region as an opportunity for agricultural development, especially cotton production. The government thought that cotton production was economically and industrially valuable, considering the suitable conditions in the region to grow it, except for the lack of rainfall. For this purpose, the Soviet Union came up with an idea of engineering the environment by bringing water to agricultural land.

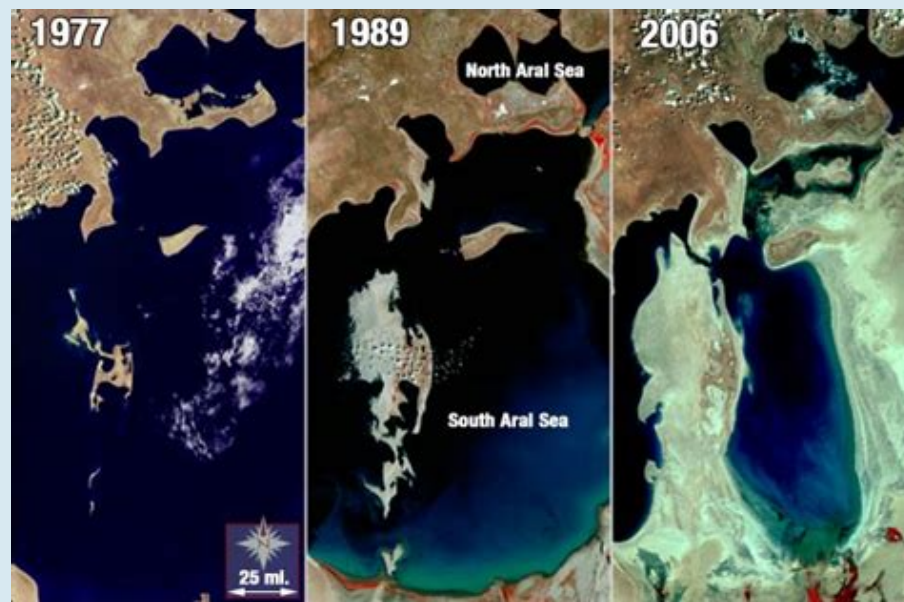
After this, the government started to construct large networks of canals, dams, reservoirs, and irrigation systems that focused on redirecting water from the region’s main rivers, the Amu Darya and Syr Darya, towards agricultural fields. The Karakum Canal was one of the mechanisms used to transport water from the Amu Darya across Turkmenistan. As previously stated, this was planned to create new agricultural land in areas that were too dry.

The 1960s was the main turning point for the project planned by the Soviets. It was an age of intensification of the irrigation project and cotton production. This converted vast areas of Central Asia into farmland. Studies made by the FAO also indicate that irrigated land increased from 4.5 million hectares to 7 million hectares between 1960 and 1980. At this point, the consequences became easily visible when the water volume in the Aral Sea started decreasing. Since the two main rivers of the region had their water redirected away from the sea, there was significantly less water flowing into the Aral Sea. With the contribution of the very dry climate in the region and the irrigation project, the water in the sea started to evaporate, causing it to become much smaller than before.

What makes this even more interesting is that the irrigation itself was inefficient. This was mainly due to the irrigation infrastructure, which was inefficient. Water was lost through leakage from canals, seepage into the surrounding soils, and evaporation. Additionally, poor drainage also created other problems. Soil salinity also started becoming concerning. In an arid environment, irrigation water contains dissolved salts.

When plants use some of the water and the rest evaporates, it leaves salts in the soil, making it saline and unsuitable for crops to grow in. Although the system was designed to keep agricultural productivity growing, it became a reason for its loss on a huge scale. Later, the Aral Sea started to shrink in volume dramatically. This increased the salinity of the sea and contributed to the loss of many aquatic animals, such as the Aral barbel and Aral nase.

The seabed also started to become exposed due to the shrinkage of the sea. It contained large quantities of salts and sediments, including agricultural chemicals, which strong winds were able to carry into the surrounding environment. It doesn't end here; it also created the Aralkum Desert. Furthermore, it impacted the fishing industry as it destroyed much of the sea's coverage.



Evolution of the Aral Sea, from 1977 to 2006

The impacts of the Soviet project don't end here. It has also resulted in major human and political problems for the region and its governments. Humanly, it affected the people who were dependent on fishing, as it was a source of employment and food. Also, the exposure of the seabed resulted in environmental and public health concerns. Politically, the two main rivers of the region, the Amu Darya and Syr Darya, were transboundary rivers that crossed many republics. This itself led to competition between the river basin countries, as they had different priorities for how to use the water. This heated up the hydropolitics of the region.



After the fall of the Soviet regime, there have been many movements to restore the water of the sea. Some of them include Kazakhstan's establishment of the Kok-Aral Dam with support from the World Bank. It was a dam that separated the North Aral Sea from the southern basin and partially restored water levels and fisheries. More recently, the countries of Central Asia have increasingly worked to manage the Amu Darya and Syr Darya through different initiatives.

“This project has been taken as an example of how improper planning on a project for just one benefit, without focusing on its drawbacks and how much it affects the environment and other dimensions of society, can lead to disastrous consequences. “

Even though the Soviet government started it as a way to grow its impact across the region through agriculture, it ended up being a major problem for the region. This teaches a great lesson on how improper implementation of certain beneficial projects can end up harming society.

AN ADAPTATIVE BIOSECURITY PROGRAMME: WHAT COULD WE DO TO MAXIMISE UTILITY WITH £100 MILLION?



By Danqi Zhang – 3RD place





INTRODUCTION

If you had £100 million to reduce the risk of the next pandemic, how would you use it? I would fund the Adaptive Biosecurity Programme, aimed at reducing the risk of a novel, bioengineered pandemic and limiting its potentially catastrophic consequences if one does occur. Pandemics – natural or engineered, accidental or deliberate – are devastating, resulting in deaths, huge economic setbacks and losses in development. However, bioengineered pandemics require particular attention as technical barriers fall and AI capabilities evolve alongside biotechnology. The threat landscape is changing faster than current biosecurity systems can adapt. Additionally, natural selection favours reproductive fitness while deliberate engineering could potentially optimise harmful features that are not primarily determined by evolution.

Although our existing infrastructure is important, most biosecurity strategies rely on data from known pathogens. New, engineered pandemics pose a risk precisely because their characteristics may be unknown.

The £100 million would be used to address gaps across the existing biosecurity chain. This would strengthen the prevention infrastructure, early detection and containment capabilities, reducing the risk of a bioengineered pandemic and mitigating its potentially catastrophic consequences.

HOW THREATS ARE EVOLVING

Lower technical barriers

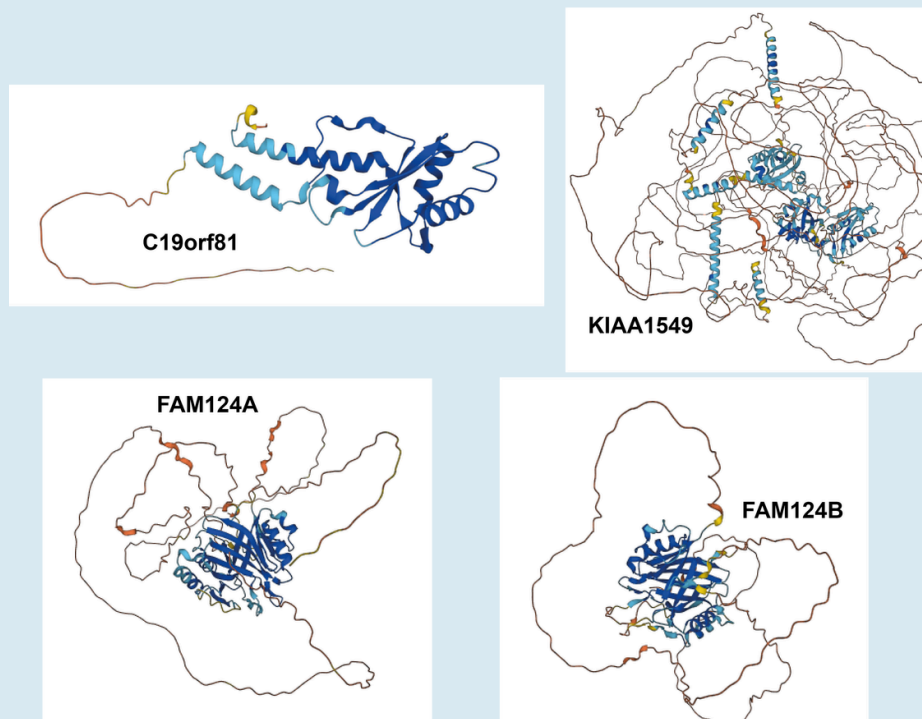
Greater accessibility to advanced genetic engineering tools, such as CRISPR-Cas9, has decreased the cost whilst improving the programmability of genome editing.

Lower technical barriers increase the number of actors capable of using complex genetic engineering. This increases the challenge for current biosecurity systems. Although progress in dual-use research is necessary to advance legitimate scientific research, we must prepare our infrastructure to prevent misuse of technology.

AI and biological information

The availability of genetic sequence data has expanded. As we collect more information about pathogens through legitimate research, more biological data is available that could be misused. Capable actors may therefore face fewer technical barriers to engineering pathogens than before.

Furthermore, the rise of AI has changed the threat landscape, through its ability to aid the gene sequencing and editing processes. AI may be able to process enormous amounts of biological information quickly, identifying patterns in previously studied pathogen genomes.



Proteins developed by AlphaFold, an AI-powered protein predictor

WHY CURRENT INFRASTRUCTURE IS NOT ENOUGH

Synthesis Gap

Preventing the actual synthesis of dangerous pathogens is a very efficient stage of containment as it drastically reduces potential harm caused. Furthermore, intervention is more effective at this phase since accurate sequence information is available for evaluation. [5] However our current DNA screening systems have important limitations and broadly depend on our existing knowledge of diseases. The standard screening processes for nucleic acid synthesis providers involves going through the customer's order and flagging any sequences of concern (which are nucleotide sequences that are known to contribute to pathogenicity, virulence or toxicity). [3] The UK guidance acknowledges the limitations of current defense mechanisms such as: the possibility of sequences being distributed across multiple orders/providers as an existing challenge. Furthermore, although many dangerous pathogens may share similarities in DNA, sequence based matching cannot perfectly determine biological function. The system must continuously evolve.

An effective screening system only works when the information used to create it is constantly updated as the biosecurity threat landscape evolves.



Early Warning Gap

Traditional pathogen surveillance is used for monitoring diseases we already recognise rather than anticipating new ones.

Additionally, although metagenomics offers a more comprehensive approach for diagnostics, since it sequences nucleic acids from all genetic material in a sample, it can eventually detect and flag even very rare genomes. [4] Unfortunately, this technology is still relatively new, less developed and limited by a lack of bioinformatics.

PREVENT, DETECT, CONTAIN

A simplified pathogen pipeline consists of: capability, synthesis, release, detection, transmission, containment. The Adaptive Biosecurity Programme aims to disrupt this chain at multiple points through the interconnected Prevent, Detect, Contain framework.

Prevention

Prevention focuses on stopping dangerous bioengineered pathogens before they are even synthesised. The investment of 25 million pounds will upgrade existing DNA screening infrastructure through integrating AI in. AI-assisted updates to the database will help ensure our screening mechanisms are always as accurate as possible.

Detection

For this phase, we target the potential release of a disease. As mentioned before, traditional surveillance can be inefficient, relying on existing data. However, by combining metagenomic screening with AI, we can improve the accuracy of this system and minimise the problems related to storing genomic data with AI. £40 million out of the total £100 million would be used here because metagenomic surveillance has the least available foundation. Furthermore, deliberate release is cited as an important way that a pandemic's damage can be maximised.[2]

Containment

Tackling transmission is the last stage of the pathogen pipeline. Since we could be dealing with unknown threats, outbreak modelling and adaptable response technologies would be useful in improving the effectiveness of our immediate countermeasures. Therefore, I would distribute £ 20 million to containment, the least amount of money for the three-part phase, as managing the spread of a pandemic is much less cost-effective than averting the potential pandemic earlier on



Self-administering COVID-19 test, developed for quick diagnosis during th 2020 pandemic.

CONCLUSION

The most pressing problem in current biosecurity is the challenge of adapting to constantly shifting pathogen capabilities . The Adaptive Biosecurity Programme would strengthen prevention, detection and containment strategies, allowing biosecurity to keep up with technologies that are changing the threat landscape.



SOLAR LANTERNS AS AN OFF-GRID ENERGY SOLUTION

BY EDWARD K. CONNOLLY - 2ND PLACE



In Nigeria, nearly half of communities in the Niger Delta receive fewer than two hours of electricity per day, despite being connected to the regional power grid (Stakeholder Democracy Network, 2016). Beyond the Delta, the problem is even more widespread: the Nigerian government estimates that more than 80 million people nationwide lack reliable access to electricity (Stakeholder Democracy Network, 2016). As a result, many households rely on kerosene lamps, diesel generators, and wood fires for light and energy, all of which release pollutants and increase the risk of respiratory illness.

This problem extends beyond Nigeria. Before 2000, nearly 40% of India's population also lacked electricity, but large-scale electrification programs, including government initiatives that distributed more than 600,000 solar lanterns to rural communities, helped reduce that figure to less than 1% (Khanna & Rowe, 2024, p. 1). Solar lanterns offer an innovative off-grid solution by converting sunlight into electricity that can be stored and used after dark (Fig. 1). Although manufacturing these lanterns' solar panels and batteries can generate electronic waste, solar lanterns have an overwhelmingly positive impact because they reduce air pollution, protect users from the health risks associated with fuel-burning lamps, and provide reliable light that allows people to work and study after sunset.

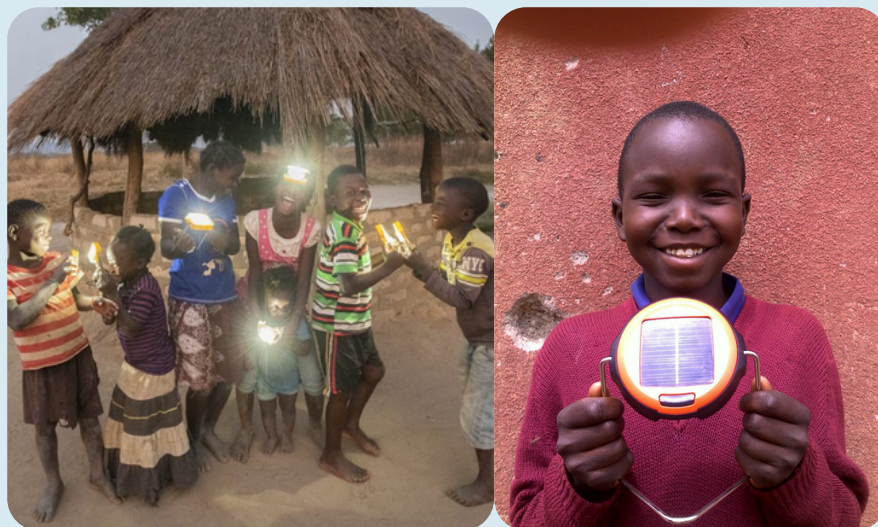


Figure 1: Children using solar lanterns in Africa. The image on the left shows a typical lantern, while the image on the right shows the photovoltaic panel that collects solar energy (Namene Solar, n.d.; Keane, 2012).

THE DEVELOPMENT AND OPERATION OF SOLAR LANTERN TECHNOLOGY:

The development of solar lanterns began with a major breakthrough in solar technology. In 1954, researchers at Bell Labs created the first practical silicon photovoltaic cell, which converted sunlight directly into electricity (Machín & Márquez, 2024, p. 2). However, solar lanterns did not become widely affordable and commercially available until later advances made solar cells less expensive and LED bulbs more energy efficient (Lighting Africa, 2013). Modern solar lanterns combine these technologies with rechargeable batteries to produce and store light without relying on a power grid (Kofoworola et al., 2025; Fernandez-Fuentes et al., 2021).

Solar lanterns generate electricity when the attached panels come into contact with photons, or particles of sunlight. As Figure 2 indicates, photons interact with the solar cells on the lantern, producing what scientists call the “photovoltaic effect” (Walker, 2026). Most photovoltaic cells are made from silicon and contain two layers: one with phosphorus to provide extra electrons, and another with boron to create spaces that electrons can move into. The boundary between these layers produces an electric field (Walker, 2026). When sunlight hits the solar cell, it causes electrons to move, creating an electric current. This electricity charges a battery, which stores the energy until it is needed to power the lantern’s bulb. Because LEDs can produce bright, long-lasting light with very little electricity, they are especially well suited for solar lanterns.

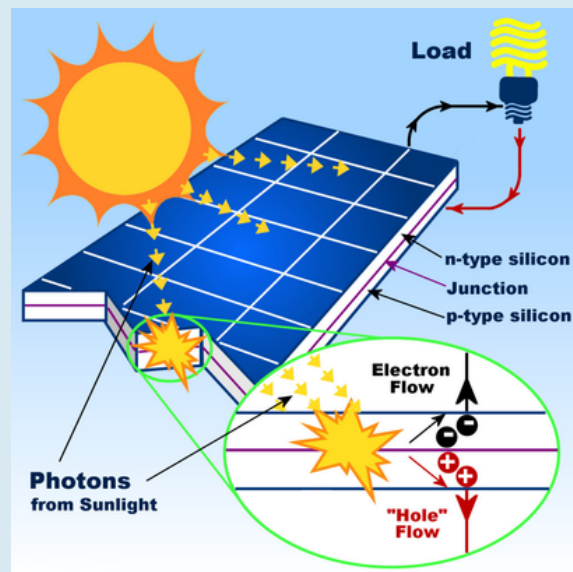


Figure 2: The process by which sunlight interacts with the phosphorus- and boron-modified layers of a silicon solar cell to generate an electric current (RPS Solar Pumps, n.d.).



THE HEALTH, ECONOMIC, AND ENVIRONMENTAL BENEFITS OF SOLAR LANTERN TECHNOLOGY

Across the world, communities that have adopted solar lanterns have benefited from healthier living conditions and increased economic productivity. Kerosene lamps produce concentrations of particulate matter well above the World Health Organization's recommended 24-hour average of 15 micrograms per cubic meter (C40 Knowledge Hub, 2021). Particulate matter measuring less than 2.5 micrometers in diameter is especially dangerous because it can reach the lungs while breathing. Particles this size can even enter the bloodstream, increasing the risk of heart disease. In Uganda, Muyanja et al. (2018) measured particulate matter concentrations ranging from 20.2 to 270.0 micrograms per cubic meter in households using kerosene lamps. After those households adopted solar lanterns, exposure decreased by 95% (Chowdhury & Naz, 2023).

Solar lanterns can also extend the number of productive hours available each day. A report examining communities in Kenya and Cambodia found that access to solar lighting made it easier for people to operate businesses and perform other work-related activities. In Kenya, nearly one-third of households using solar lanterns experienced an economic benefit. In Cambodia, 37% of respondents reported earning additional income because they could work later into the evening, producing an average 18% increase in earnings among those households (Lemaire, 2018). The lantern's portability and independence from a central grid are therefore crucial parts of this scientific breakthrough: electricity can reach households and businesses without governments needing to construct costly transmission infrastructure.

Solar lanterns have a substantially smaller environmental footprint than kerosene lamps. Burning kerosene releases carbon monoxide, sulfur and nitrogen oxides, and black carbon (Lam et al., 2012). Black carbon is particularly damaging because it absorbs sunlight and rapidly warms the atmosphere. Research indicates that, although black carbon remains in the atmosphere for less time than carbon dioxide, it produces an especially intense warming effect (Tesden, 2013, p. 3). Because off-grid lighting remains common across South Asia and sub-Saharan Africa, replacing fuel-burning lamps with solar alternatives could help reduce the approximately 270,000 metric tons of black carbon produced by kerosene lighting worldwide each year (Tesden, 2013, p. 3).



SMALL-SCALE INNOVATION, GLOBAL BENEFITS

It is important to note that solar lanterns cannot fully compensate for unreliable electrical grids or provide enough energy to power entire homes or businesses. Still, their overall net impact is clearly positive. By combining silicon photovoltaic cells, rechargeable batteries, and energy-efficient LEDs in one portable device, engineers transformed several earlier scientific advances into a solution for inadequate access to electricity. In Nigeria, India, Cambodia, and Uganda, solar lanterns reduced exposure to hazardous particles, lowered carbon emissions, and created the infrastructure that people need to work safely and more productively. The success of solar lanterns in these countries demonstrates how a relatively small technology can have significant health, economic, and environmental benefits, especially in underserved regions. Studying the history and engineering behind solar lanterns suggests that some of the greatest scientific innovations stem from developing alternatives to previous models in the service of building a healthier, more sustainable future for more communities.



A SERENDIPITOUS SPILL: THE CHEMISTRY BEHIND SWEETNESS

BY EKA KRISHNAN - Winner





INTRODUCTION

In June 1878, chemist Constantin Fahlberg sat down to dinner after a long day in Ira Remsen's laboratory at Johns Hopkins University (Hicks, 2010). Reaching for a bread roll, he noticed an intensely sweet flavor on its crust. Realizing he had forgotten to wash his hands after working with coal tar derivatives, Fahlberg rushed back to the lab to taste the contents of every beaker and dish on his bench. He found the culprit: benzoic sulfonide, later branded as saccharin, or artificial sugar (Hicks, 2010).

That accidental spill opened the door to new research on artificial sweeteners. Saccharin turned out to be roughly 300 times sweeter than regular cane sugar, yet it carried zero calories (National Cancer Institute, 2023; Hicks, 2010). Over the next century, the U.S. Food and Drug Administration (FDA) would go on to approve several other zero-calorie sugar substitutes, including aspartame, sucralose, and acesulfame potassium (U.S. Food and Drug Administration, 2023).

Thanks to Fahlberg's accidental discovery, scientists have found a way to let us enjoy the sweet taste we naturally crave without any of the health drawbacks of sugar. But like so many other major scientific breakthroughs, solving one problem ended up creating another. While artificial sweeteners succeeded in cutting down sugar, they also paved the way for something far more troubling: the explosion of ultra-processed foods in our everyday diets.

HOW THE BODY RECOGNIZES SWEETNESS

To really understand how these sweeteners changed the way we eat, it helps to look at how our bodies process taste in the first place. When we eat something sweet, tiny receptors on our tongue act like lock-and-key mechanisms. Regular sugar fits into these receptors, sending a quick signal to our brain that says, "Hey, this tastes great, and it's full of energy!" Two specific proteins, called TIR2 and TIR3, act as our taste receptors for sweetness during this process, as indicated in Figure 1 below (Fernstrom et al., 2012).

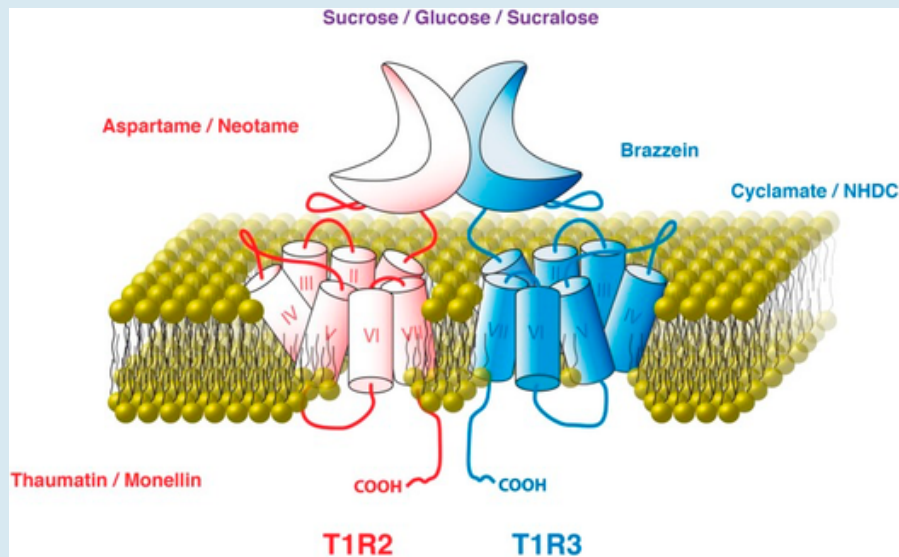


Figure 1: A diagram of how artificial sweeteners like aspartame activate the T1R2 and T1R3 proteins, humans' main taste receptors, compared to natural sugars like sucrose. Figure adapted from Fernstrom et al., 2012, p. 1135.

Artificial sweeteners are engineered to fit into those exact same receptor locks, but with two clever adjustments. First, synthetic compounds bind to our taste receptors far more tightly than natural sugar does. Because of that extra strength, food companies only need to use tiny amounts, sometimes up to tens of thousands of times less than real sugar, to give us that same sweet kick (National Cancer Institute, 2023). Second, compounds like saccharin are incredibly stable. Our digestive systems simply can't break down their unique chemical bonds (del Pozo et al., 2022, p. 12). Since our bodies can't digest them, they pass right through us unchanged. On the one hand, this means that the body cannot break these compounds down and use them for energy. On the other hand, these chemical properties meant that people living with diabetes or those trying to manage their sugar intake could enjoy sweet foods without worrying about dangerous spikes in blood sugar.

If artificial sweeteners were such a great solution for reducing sugar, where did things go wrong? The problem isn't necessarily the sweetener itself, but how the food industry began using it.

In traditional baking and cooking, real sugar does a lot more than just make items taste good. It adds bulk, texture, moisture, and structure to food. When you take sugar out, the food looks and tastes pretty unappealing. High-intensity sweeteners offered a sneaky workaround: manufacturers could use cheap chemical fillers, water, and industrial emulsifiers to rebuild the food's structure, and then add a tiny pinch of artificial sweetener to fix the flavor (Seto et al., 2025, p. 2).



Rather than making simple foods healthier, artificial sweeteners became the main building block for ultra-processed foods. Recent nutrition research shows that diet drinks and low-calorie snacks almost never rely on just one simple sweetener anymore. Instead, they contain complex mixtures of multiple additives (Sylvetsky et al., 2016).

Whole Foods vs. Ultra-Processed Foods and Their Effect on the Body

Characteristic	Natural Foods	Ultra-Processed Sweetened Products
Source of Sweetness	Simple sucrose or fructose	Combined aspartame, Acesulfame potassium, or sucralose
Ingredients	Whole, recognizable food bases	Added preservatives, thickeners, and flavor boosters
Signals	Sweet taste matches incoming energy	Sweet taste with no energy
Eating Pattern	Helps body feel full naturally	Designed to keep you reaching for more

Table 1: Table indicating how the source of sweetness, ingredients, key signals, and eating patterns of natural foods vs. ultra-processed foods impacts energy levels and satiety

ENGINEERING FOOD FOR HEALTH: IS IT WORTH IT?

When we weigh the pros and cons of artificial sweeteners, it becomes clear that the story isn't purely black or white. Because artificial sweeteners are so overwhelmingly sweet, eating them regularly can actually change how our taste buds and brains react to food (Northwestern Medicine, 2025). Over time, naturally sweet foods like fresh apples or berries start to taste plain and boring, driving us right back toward processed snacks. Still, artificial sweeteners offer undeniable short-term perks. Clinical research dietitians note that non-nutritive sweeteners can genuinely help people with Type 2 diabetes keep their glycemic levels under control.

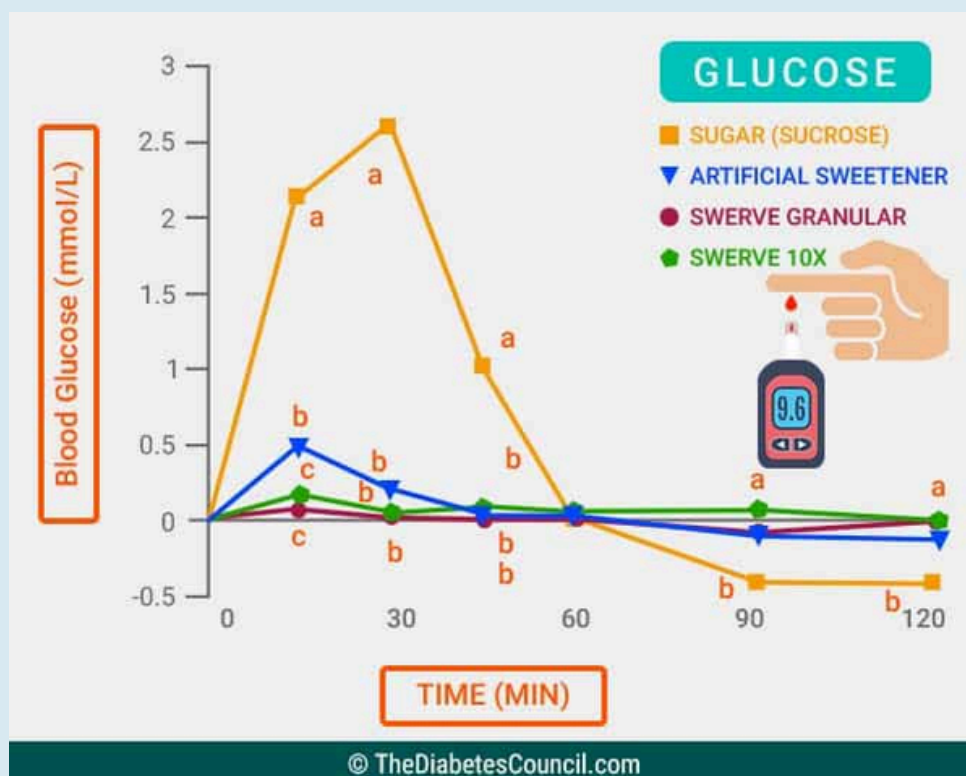


Figure 2: Graph indicating how sugar and artificial sweeteners impact blood glucose levels over time. Adapted from Manderfeld, 2020.

Science has come a long way since Constantin Fahlberg's accidental discovery of benzoic sulfonide in 1878. Chemists can now test multiple compounds at scale and in different combinations to improve a variety of human health conditions, from creating substitutes for ingredients that exacerbate chronic illnesses or developing alternatives for common allergens. Examples of such innovations include salt substitutes that lower sodium intake and reduce blood sugar (Miranda, 2025, p. 1). While we need to be careful altering one part of a food through chemistry because doing so can change how the body responds to an otherwise natural ingredient, engineered foods offer significant benefits for reducing certain health risks when we understand how they interact with the body as a whole.

“Chemistry will continue to offer us greater control over the foods we can create and consume, allowing us to preserve the qualities people enjoy while making food safer and more accessible to people with different health needs. “



THIS MONTH'S OPPURTUNITIES

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NON-TRIVIAL

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FURTHER READING

& KEY SOURCES

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- [Navier-Stokes Equations - Numberphile](#)

PILOTING A BROKEN ROCKET

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ADAPTATIVE BIOSECURITY PROGRAM

- <https://80000hours.org/problem-profiles/preventing-catastrophic-pandemics/full-report/>
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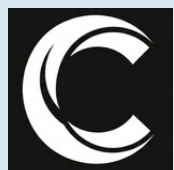
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